Expressions and operator order

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

TEACHER GUIDE · 1 OF 9

Prepare the purpose, prerequisites and materials

> Prepare Expressions and operator order: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Evaluate numeric **expression**s using Python's grouping rules

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Values variables and types

Without opening the old lesson, explain one key idea from Values variables and types. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish integer, floating-point, text and Boolean values

Reference example, not the only acceptable wording: 12 is int; 1.5 is float; "12" is str; False is bool; "False" is str. Quotes mark text even when the characters spell a Boolean name.

Teach grouping through meaning before mnemonic rules. Multiplication and division share **precedence**; addition and subtraction share **precedence**.

This lesson deliberately limits whole-group examples to nonnegative integers and positive divisors.

Reference: Python tutorial arithmetic, https://docs.python.org/3/tutorial/introduction.html, and **expression precedence**, https://docs.python.org/3/reference/**expression**s.html#**operator-precedence**.

Outcomes

Evaluate numeric **expression**s using Python’s grouping rules

Use **parentheses** to express the intended calculation

Distinguish ordinary division, whole groups and a **remainder**

Preparation

Python 3 editor with **expression**s.py and separate source/**output** areas

Counters or sketches of packs with separate loose items

Blank **expression**, grouping, expected and observed columns

Prerequisite answer

Numeric addition gives 10; text concatenation gives the string 82. This week uses numeric values.

End goal for the pictorial case

Trace **expression**s and **operator** order and explain the changed program before running it.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain and model the mechanism

> Model Expressions and operator order using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Explain Expressions and operator order · 1

Three packs each contain eight counters, with two more loose counters. First count the packed counters, then add the loose ones. A program needs that structure, not just the right numbers.

This week you will predict **expression**s, make grouping visible and check division against a physical interpretation.

Explain Expressions and operator order · 2

Use numeric values in a fresh Python 3 script called **expression**s.py. Keep expected and observed results separate as before. Multiplication and division group more tightly than addition and subtraction.

Parentheses let you state a different grouping. A correct calculation still depends on correct quantities and units.

The script displays 26. Replacing the total **expression** with packs * (per_pack + loose) displays 30, because it adds two to every pack. **Parentheses** are not decoration: they can change the represented situation.

Write a brief phrase for each group before running.

Term	Meaning
Expression	Code evaluated to produce a value
Operator	A symbol such as + or * that specifies an operation
Precedence	Rules that determine how an expression is grouped
Parentheses	Symbols that make a grouping explicit
Remainder	The amount left after forming whole groups

Two expressions with different meanings

EXPRESSION	FIRST CALCULATION	FINAL VALUE
3 * 8 + 2	3 * 8 is 24	24 + 2 is 26
3 * (8 + 2)	8 + 2 is 10	3 * 10 is 30
(3 * 8) + 2	3 * 8 is 24	24 + 2 is 26

Adding two inside every pack differs from adding two loose counters once.

In 18 - 4 * 3 multiplication groups first, giving 18 - 12 = 6. In (18 - 4) * 3 the **parentheses** give 14 * 3 = 42.

Exit: 23 // 4 gives 5 trays and 23 % 4 gives 3 tiles; 5 * 4 + 3 = 23. / gives 5.75, a ratio rather than both requested whole-number counts.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

```
print(20 - 6 - 2)
print(20 - (6 - 2))
print(12 / 3 * 2)
```

Model the reasoning aloud · 3

The first line groups subtraction from left to right: (20 - 6) - 2 gives 12. The second gives 16. Multiplication and division share a **precedence** level, so 12 / 3 * 2 groups as (12 / 3) * 2 and displays 8.0, not 2.0.

Ordinary / division produces a float, even when the result is mathematically whole.

> **Explain why 18 - 4 * 3 gives 6 while (18 - 4) * 3 gives 42. Which operation is grouped first in each expression?**

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Expressions and operator order

Parentheses change an **expression**. A toy shelf calculator compares adding a fixed cost after multiplication with multiplying an entire group.

The facts we will use

> **Supplied facts**
>
> Python 3: print(3+2*5); print((3+2)*5). **Input**s are dimensionless for this arithmetic exercise.

AI ACADEMY / PYTHON-BRIDGE/3

Expressions and operator order

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Use precedence

In 3+2*5, multiplication happens first.

2. Compute first result

2*5 is 10; adding 3 gives 13.

3. Compute grouped result

Parentheses give 3+2=5 first; multiplying by 5 gives 25.

CHANGE ONE THING

Replace the final multiplier 5 with 4 in both expressions.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Use precedence	In 3+2*5, multiplication happens first.
2. Compute first result	2*5 is 10; adding 3 gives 13.
3. Compute grouped result	Parentheses give 3+2=5 first; multiplying by 5 gives 25.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Parentheses explicitly group an operation. Without them, Python multiplication takes priority over addition; the same numerals can therefore describe different procedures.

What this example does not establish

Keep the stated limit

Arithmetic correctness does not decide which **expression** matches a real problem; the written model must define the intended quantities.

Change one thing in the pictured case

Change one thing

Change: Replace the final multiplier 5 with 4 in both **expression**s.

Predicted result

Prediction: The first result is 11; the grouped result is 20.

Why the result follows

Reason: 3+2*4=3+8; (3+2)*4=5*4.

Ask while showing the picture

Ask while showing the picture · 4

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Expressions and operator order, reduce support deliberately and check individual reasoning before discussing answers.

Choose the pacing route

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve numeric addition versus text joining.
7 to 20 minutes	Model packs and loose counters with the diagram; complete W1.
20 to 33 minutes	Read equal-precedence expressions from left to right and predict W2.
33 to 45 minutes	Run W2, compare results and collect the guided explanation.
Session 2 start to 8 minutes	Revisit one grouping disagreement with counters or brackets.
8 to 20 minutes	Model 17 counters in groups of five and complete W3.
20 to 35 minutes	Collect W4 independently, including the reconstruction check.
35 to 45 minutes	Complete W5 and the exit, recording input assumptions and units.

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Colour the pack calculation and loose quantity separately, then use the same grouping in code. Provide an operation legend without doing W4.

Extension: devise two different **expression**s that agree for one **input** and differ for another, and explain the testing consequence.

Reduce help deliberately

Completion practice

Completion task: Predict 16 - 5 - 3, 16 - (5 - 3), and 18 / 3 * 2. Write the grouping, run each

line and record actual results separately.

Reasoning to listen for

Teacher check: (16 - 5) - 3 is 8; 16 - (5 - 3) is 14; (18 / 3) * 2 is 12.0. Record any observed discrepancy and check the saved **expression** rather than copying expected results.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Expression**s and **operator** order. Proposed explanation: “All operations are read left to right regardless of **precedence**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: Establish grouping first; only operations at the same applicable **precedence** level group left to right here.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Trace the operation order in both **expression**s. Identify which picture step changes with **parentheses** and predict both results when the multiplier becomes four.

Reasoned teacher answer

Without grouping, multiply before adding: 13. With grouping, add before multiplying: 25. Changing the multiplier to 4 gives 11 and 20 respectively.

Changed-case reasoning: The first result is 11; the grouped result is 20. 3+2*4=3+8; (3+2)*4=5*4.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the investigation and diagnose misconceptions

> Run the Expressions and operator order investigation and use the first unsupported step to choose a teaching response.

All operations are read left to right regardless of **precedence**: Establish grouping first; only operations at the same applicable **precedence** level group left to right here.

Parentheses never change a value: Compare the two packs **expression**s and their different stories.

Ordinary division always returns an integer: / produces a float for these integer operands.

A **remainder** can be as large as the group size: For this positive whole-group case, another complete group would then be possible.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

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Assess and explain the answers

> Use the Expressions and operator order questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Guided and exit questions

In 18 - 4 * 3 multiplication groups first, giving 18 - 12 = 6. In (18 - 4) * 3 the **parentheses** give 14 * 3 = 42.

Exit: 23 // 4 gives 5 trays and 23 % 4 gives 3 tiles; 5 * 4 + 3 = 23. / gives 5.75, a ratio rather than both requested whole-number counts.

W1 State the intended grouping

Workbook question · W1

Write an **expression** for three packs of eight counters plus two loose counters. Explain why adding two within the **parentheses** for every pack describes a different situation.

Teacher reasoning · W1

3 * 8 + 2 or (3 * 8) + 2 gives 26. 3 * (8 + 2) gives 30 and adds two to each of three packs, not two once.

W2 Predict then test expressions

Workbook question · W2

Predict 16 - 5 - 3, 16 - (5 - 3), and 18 / 3 * 2. Write the grouping, run each line and record actual results separately.

Teacher reasoning · W2

(16 - 5) - 3 is 8; 16 - (5 - 3) is 14; (18 / 3) * 2 is 12.0. Record any observed discrepancy and check the saved **expression** rather than copying expected results.

W3 Explain whole groups

Workbook question · W3

For 17 counters in groups of 5, explain /, // and %. Show a multiplication-and-addition check

for the complete groups and **remainder**.

Teacher reasoning · W3

/ gives 3.4; // gives 3; % gives 2. The reconstruction is 3 * 5 + 2 = 17, with **remainder** 2 less than group size 5. Only three complete groups can be made.

W4 Translate a new situation independently

Workbook question · W4

Four trays hold six tiles each and five tiles are loose. Write and evaluate the total **expression**. Then pack the total into complete groups of four and find the **remainder**. Explain why 4 * (6 + 5) is wrong for this story.

Teacher reasoning · W4

4 * 6 + 5 gives 29 tiles; 29 // 4 gives 7 groups and 29 % 4 gives 1 left. 7 * 4 + 1 reconstructs 29. 4 * (6 + 5) gives 44, adding five to every tray instead of once.

W5 Check a limit

Workbook question · W5

Explain why a group size of zero must be excluded. State the assumptions that let complete groups and a **remainder** describe the packing task.

Teacher reasoning · W5

A zero-size group cannot define how many groups consume the counters; Python division by zero raises ZeroDivisionError.

The physical task uses whole nonnegative counters and a positive whole group size, so the **remainder** is between zero and one less than group size.

Independent assessment and fresh retry

Teacher key · Independent assessment and fresh retry

W4 needs 29, 7 and 1 with the meaning of the extra five and a reconstruction. After support, retry three trays of seven plus four loose: 25 total; groups of six give 4 complete groups and 1 left.

Require an **expression** and explanation before a device result.

Assess independent transfer

Independent question · withhold the key

Selected case: Predict each line. Explain which calculation groups addition before division. What would (17+5)%5 return, and how do /, // and % differ for the supplied positive integers?

```
print((17 + 5) // 5)
print(17 + 5 // 5)
print(17 / 5, 17 // 5, 17 % 5)
```

Selected case answer

Teacher key · Selected case answer

The first line is4 because22//5=4. The second is18 because5//5 is evaluated before adding17. The final line is3.4 3 2: ordinary division, whole quotient and **remainder** for17 divided by5. (17+5)%5 is2.

Parentheses choose the grouped calculation; they are not optional when that meaning is intended.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect the project, support and next lesson

> Connect Expressions and operator order to Python readiness portfolio, adapt support and plan a later changed-case check.

Add a labelled **expression**s and **operator** order **evidence** sheet to Python readiness portfolio. Show how you can evaluate numeric **expression**s using python's grouping rules.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Colour the pack calculation and loose quantity separately, then use the same grouping in code. Provide an operation legend without doing W4.

Extension: devise two different **expression**s that agree for one **input** and differ for another, and explain the testing consequence.

Week 4 introduces comparisons and Boolean conditions that express whether requirements hold.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: python-bridge/4, python-bridge/6, python-bridge/9. Use actual classroom dates, adjusting for holidays and gaps.

One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

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Prepare, teach and assess

> Use this preparation summary alongside the detailed Expressions and operator order lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Evaluate numeric **expression**s using Python's grouping rules

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: Predict 16 - 5 - 3, 16 - (5 - 3), and 18 / 3 * 2. Write the grouping, run each line and record actual results separately.

Supported practice answer

Reasoned practice answer: (16 - 5) - 3 is 8; 16 - (5 - 3) is 14; (18 / 3) * 2 is 12.0. Record any observed discrepancy and check the saved **expression** rather than copying expected results.

Check every learner before discussion: Use the worked case for **Expression**s and **operator** order. Proposed explanation: “All operations are read left to right regardless of **precedence**”.

Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Establish grouping first; only operations at the same applicable **precedence** level group left to right here.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Replace the final multiplier 5 with 4 in both **expression**s. Require a **prediction** and a reason before comparing. Expected result: The first result is 11; the grouped result is 20. Reason: 3+2*4=3+8; (3+2)*4=5*4.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Python readiness portfolio. Keep runnable code, **prediction**s, actual **output**s, one repaired error and an explanation of a function on changed **input**s.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

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Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

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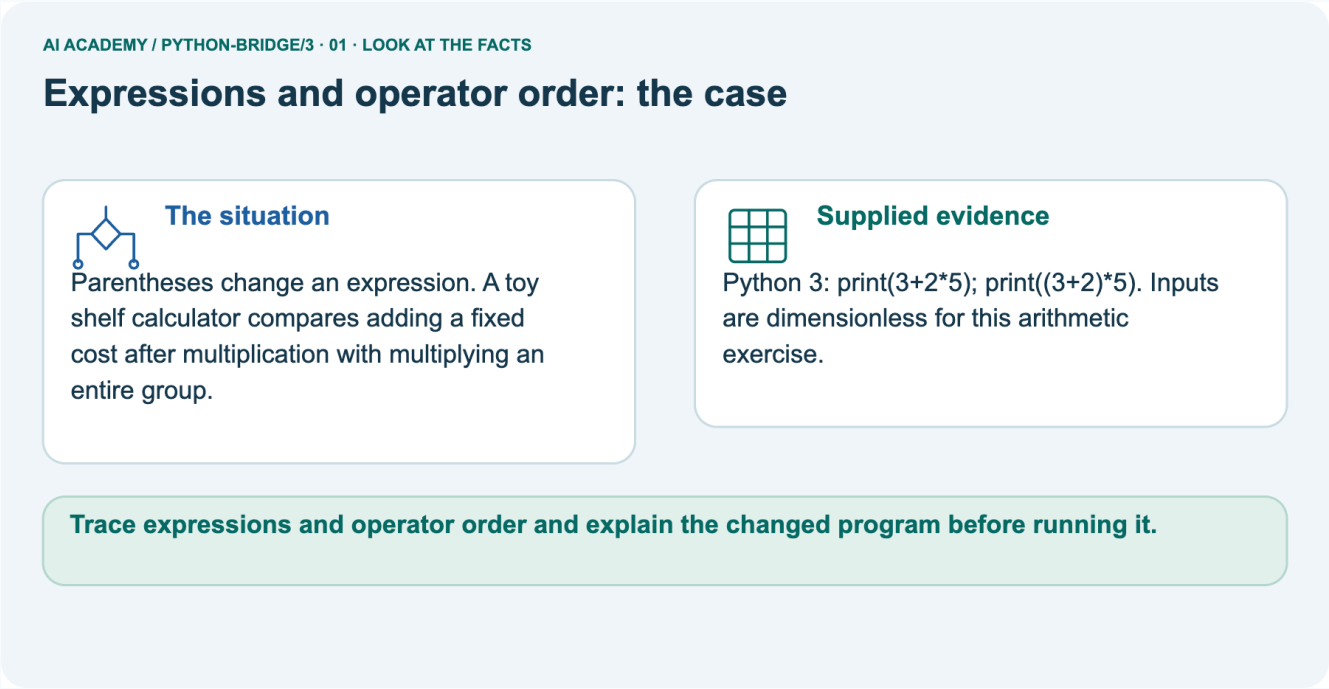
Visual concept journey

> Show one Expressions and operator order graphic at a time, request a prediction and discuss

> its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Expressions and operator order: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Parentheses change an **expression**. A toy shelf calculator compares adding a fixed cost after multiplication with multiplying an entire group. Supplied facts: Python 3: print(3+2*5); print((3+2)*5).

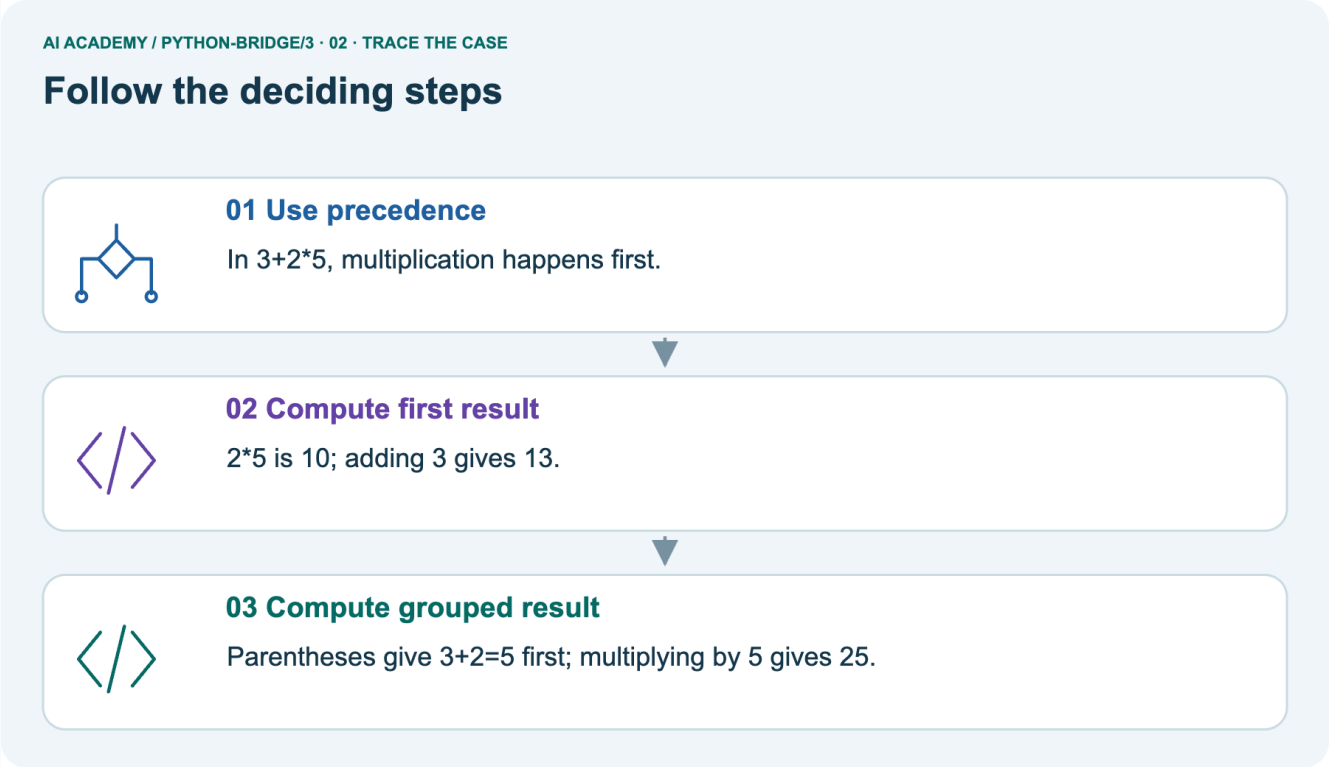
Inputs are dimensionless for this arithmetic exercise.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

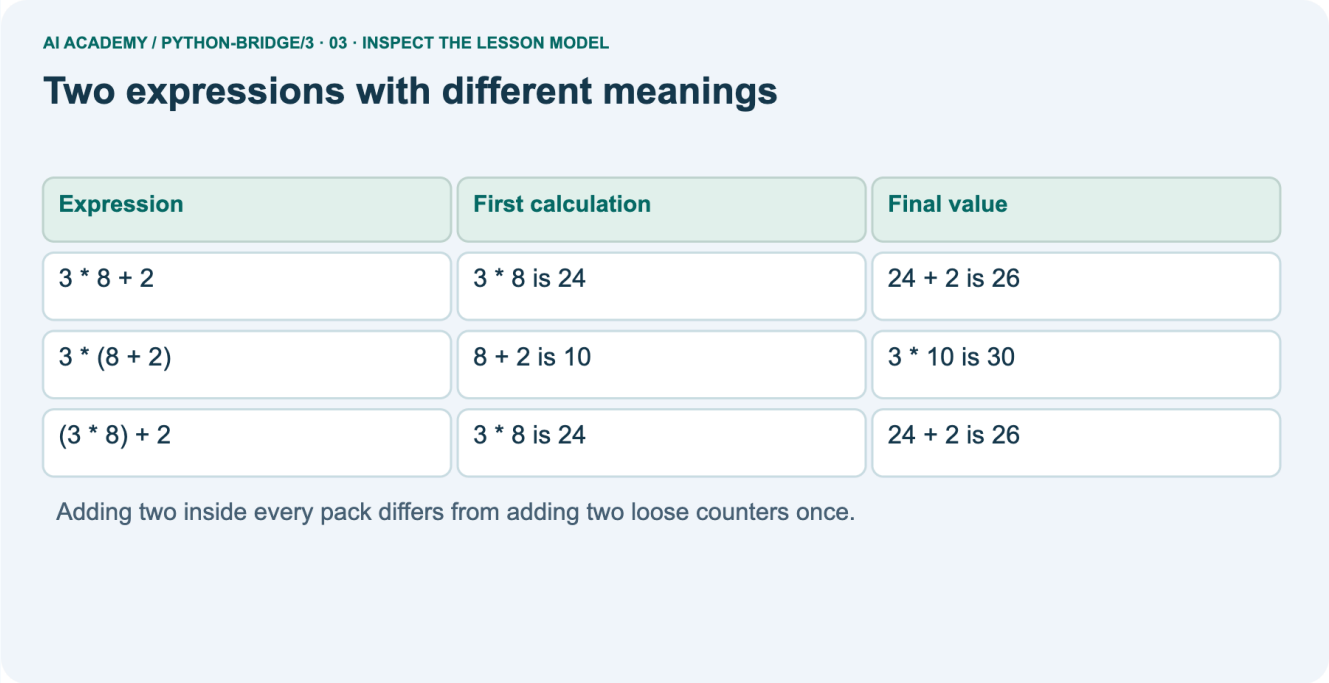
Use **precedence**: In 3+2*5, multiplication happens first. → Compute first result: 2*5 is 10; adding 3 gives 13. → Compute grouped result: **Parentheses** give 3+2=5 first; multiplying by 5 gives 25.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Two expressions with different meanings



Follow the labelled picture with the matching explanation. Open the vector diagram

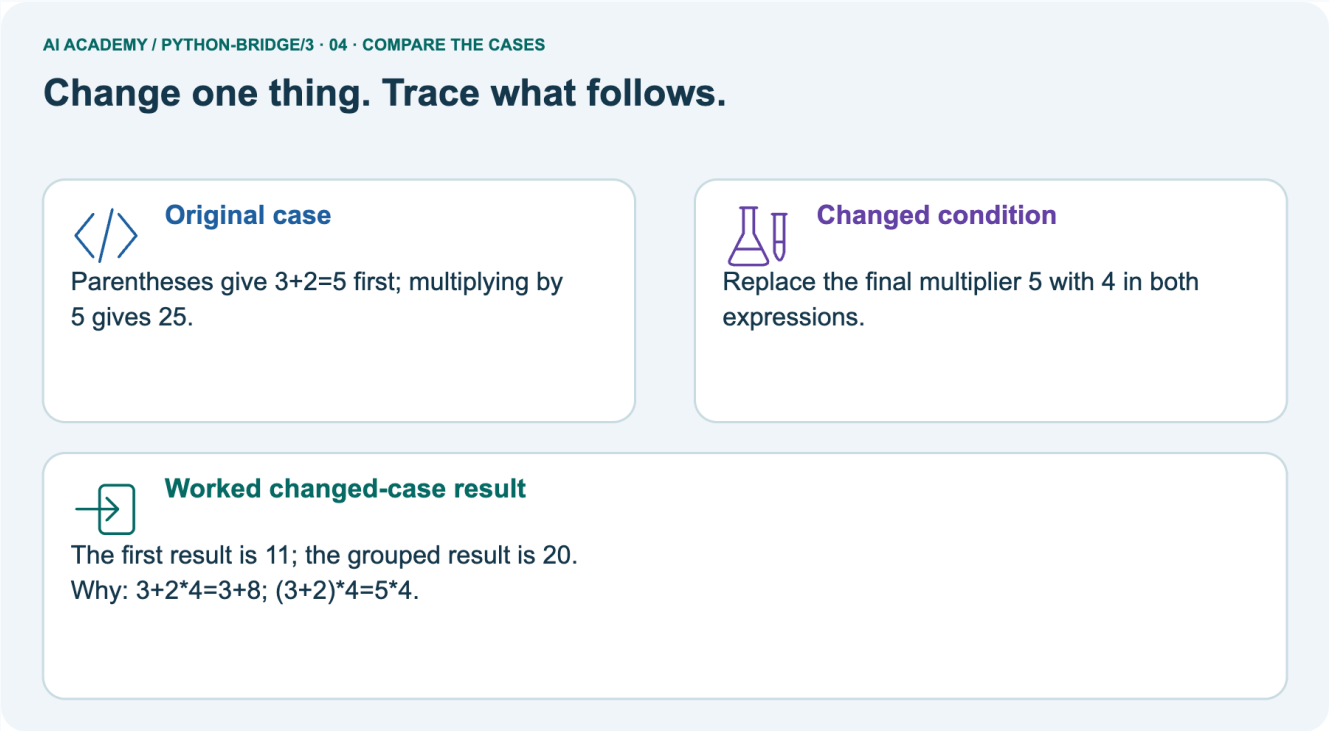
Two **expression**s with different meanings. 3 * 8 + 2 | 3 * 8 is 24 | 24 + 2 is 26. 3 * (8 + 2) | 8 + 2 is 10 | 3 * 10 is 30. (3 * 8) + 2 | 3 * 8 is 24 | 24 + 2 is 26

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: **Parentheses** give 3+2=5 first; multiplying by 5 gives 25. Changed condition: Replace the final multiplier 5 with 4 in both **expression**s. Expected result: The first result is 11; the grouped result is 20.

Reason: 3+2*4=3+8; (3+2)*4=5*4.

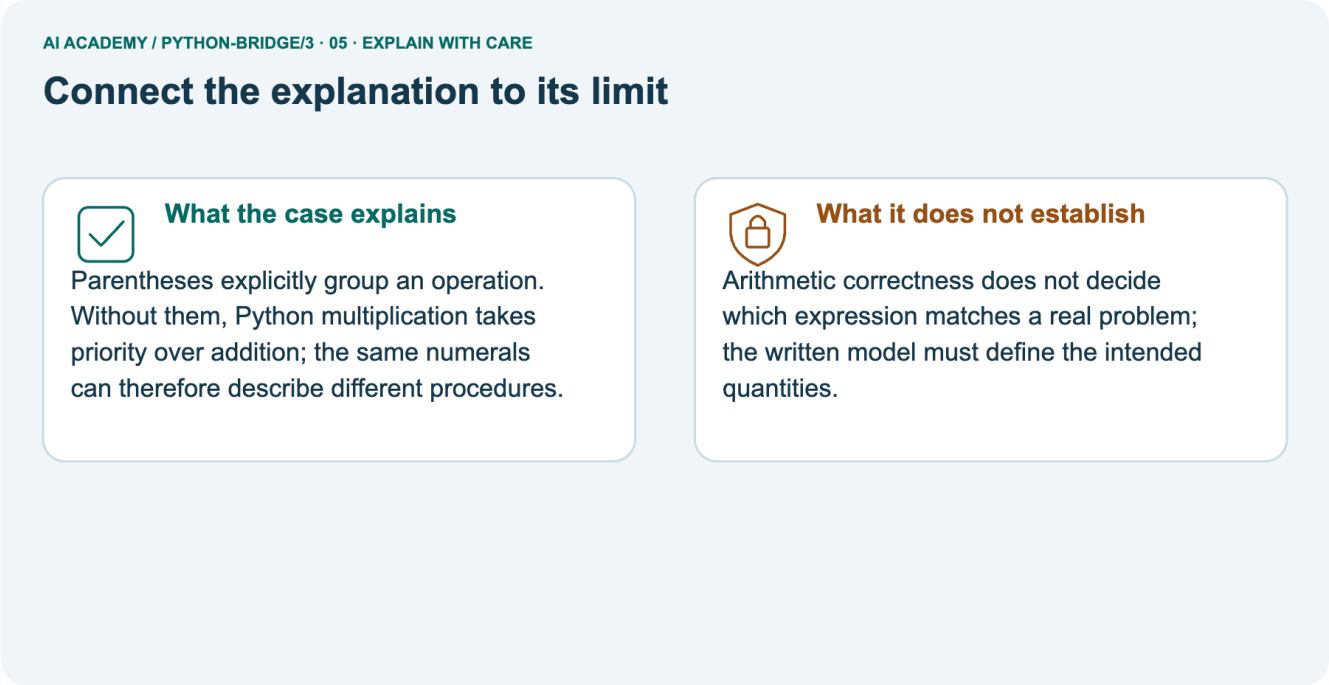
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: 3+2*4=3+8; (3+2)*4=5*4.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Parentheses explicitly group an operation. Without them, Python multiplication takes priority over addition; the same numerals can therefore describe different procedures.

Limit: Arithmetic correctness does not decide which **expression** matches a real problem; the written model must define the intended quantities.

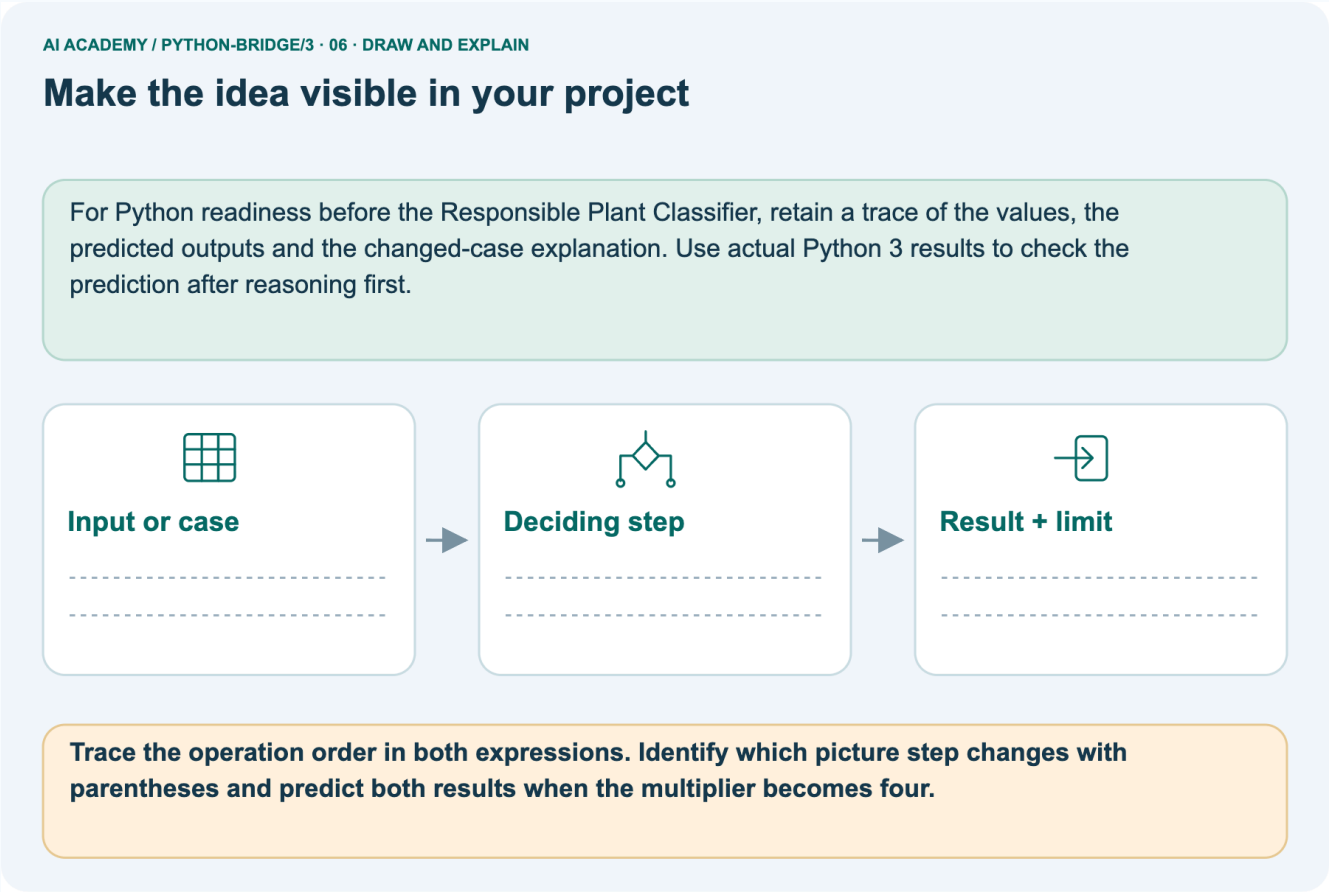
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Arithmetic correctness does not decide which **expression** matches a real problem; the written model must define the intended quantities.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: Trace the operation order in both **expression**s.

Identify which picture step changes with **parentheses** and predict both results when the multiplier becomes four.

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Without grouping, multiply before adding: 13. With grouping, add before multiplying: 25. Changing the multiplier to 4 gives 11 and 20 respectively.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.